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SUMMARY

Data and AI Engineer with production experience designing ETL/ELT pipelines, LLM-orchestrated workflows, and ML systems that eliminate manual bottlenecks and drive operational decisions at scale. Engineered an agentic AI screening platform, collapsing a 5-stage hiring process to under 2 minutes and reducing reporting latency 28% across 2M+ retail transactions at Dollar General (Fortune 100). Core stack: Python, SQL, Apache Spark, Airflow, LLM orchestration, agentic AI, PostgreSQL, FastAPI, Docker, and AWS.

SKILLS

Agentic AI & LLM: n8n, Groq API, Ollama, LLM Orchestration, RAG, Prompt Engineering, Whisper (OpenAI), Pinecone, Weaviate, Langchain, Langgraph
Data Engineering: ETL/ELT, Apache Spark, Apache Airflow, Apache Kafka, Hadoop (HDFS/MapReduce), dbt, Databricks, Data Modeling
Programming: Python, TypeScript, SQL (Advanced), PySpark, FastAPI, Flask, Node.js, Express.js, Pandas, NumPy, scikit-learn, PyTorch, XGBoost, R
Computer Vision & AI Services: OpenCV, DeepFace, MediaPipe
Databases & Cloud: PostgreSQL, Redis, MySQL, MongoDB, Snowflake, AWS (S3, Redshift, CloudWatch), Azure Synapse Analytics, GCP, Kubernetes
Tools & Platforms: Docker, Terraform, Git, GitHub, CI/CD, REST APIs, Tableau, Power BI

EXPERIENCE

Cloud BC Labs, Reston, Virginia: Software Engineer Intern — Data & AI Engineering

Oct 2025 – May 2026

- **Automated** the full candidate assessment lifecycle using n8n agentic workflow and Groq LLM; job description submission triggers role-specific question generation and candidate email delivery with no manual steps, collapsing a 5-stage hiring process to under 2 minutes end-to-end.
- **Designed** 6 PostgreSQL schema migrations from scratch — normalizing candidates, assessments, AI scores, proctoring flags, and job requisitions into a structured schema powering real-time recruiter dashboard queries for verdict tracking and pipeline reporting.
- **Integrated** 4 AI services — DeepFace for identity verification, Groq LLM for MCQ scoring, Groq Whisper for video transcription, and MediaPipe for behavioral proctoring — into a 6-container Docker pipeline with no manual routing between stages.
- **Deployed** RESTful APIs in FastAPI and Node.js, connecting the React HR dashboard to backend AI scoring services; standardized data contracts across 4 microservices, cutting cross-service integration overhead by 20%.
- **Instrumented** an end-to-end screening funnel with usage analytics; surfaced recurring failure patterns that cut incoming bug report volume 15% and shifted sprint priorities toward the highest-impact fixes.

HCLTech Ltd, Noida, India: Data Engineer — Dollar General (Fortune 100 Retailer)

Feb 2023 – Jul 2024

- **Implemented** Spark and Airflow data pipelines with partition pruning and query caching, cutting operational reporting latency 28% across Dollar General's retail analytics stack serving 19,000+ store locations.
- **Validated** 2M+ retail transaction records via Python and SQL anomaly detection, resolving data quality gaps that were degrading inventory replenishment model accuracy across Dollar General's fulfillment network.
- **Constructed** an anomaly detection and downtime forecasting model using scikit-learn and PyTorch on AWS system performance logs, cutting incident response time by 25% by flagging at-risk systems before failures reached production.

PROJECTS

Credit Risk Modeling & Default Prediction Pipeline | Python · Scikit-learn · XGBoost · SMOTE

- **Applied** SMOTE to correct a 1:7.6 class imbalance across 255K+ lending records, then benchmarked logistic regression, random forest, and XGBoost via 5-fold cross-validation—XGBoost topped the comparison at 88.5% accuracy and 0.737 ROC-AUC.
- **Compared** three classifiers by engineering 6 predictors and measuring precision-recall tradeoffs across each model, proposing a hybrid deployment strategy designed to catch high-risk defaults without over-flagging creditworthy borrowers in production.

Scalable Fleet Risk Analytics Pipeline | Hadoop · HDFS · MapReduce · Power BI

- **Architected** a distributed data ingestion pipeline using Hadoop HDFS and MapReduce to process operational datasets across 300+ assets in 14 states, reducing manual data prep time and enabling repeatable pipeline runs across any fleet expansion.
- **Developed** Power BI dashboards on the cleaned 300+ asset fleet dataset, surfacing risk scores and maintenance flags, enabling operations teams to shift from reactive incident response to predictive maintenance scheduling.

Instacart Customer Behavior Dashboard | SQL · Tableau · Python

- **Processed** 3M+ grocery transaction records through optimized SQL pipelines—developing customer segmentation logic, cohort retention queries, and demand forecasting models that fed directly into operational reporting for merchandising and fulfillment planning teams.
- **Produced** Tableau dashboards tracking supply chain KPIs, including product demand velocity, fulfillment rate trends, and campaign ROI—giving merchandising and operations teams a shared view for inventory planning and replenishment decisions.

CardioRisk — Predictive Health Analytics Pipeline | R · Statistical Modeling · ML Pipeline

- **Evaluated** 5 classification models on 319K+ CDC health records—Neural Network achieved top performance at 91.4% accuracy, 0.81 AUC, and 99.97% recall; surfaced Decision Tree overfitting by flagging its matching accuracy against a 0.50 AUC, a critical signal for clinical deployment readiness.
- **Ranked** BMI, age category, sleep time, and general health as top predictors from 17 clinical attributes via Random Forest Gini importance-adjusted classification threshold to maximize recall and minimize missed diagnoses.

EDUCATION

The University of Texas at Dallas, Texas (GPA: 3.88/4.0)

M.S., Information Technology & Management — Dean's Excellence Scholar • Grad Certificate in Business Analytics & Data Mining

USA

Aug 2024 – May 2026

Savitribai Phule Pune University, Pune (GPA: 3.80/4.0)

B.E., Electrical Engineering

India

Aug 2018 – May 2022